MIL-E-1/1653A(NAVY)
4 November 1970
SUPERSEDING
MIL-E-1/1653(NAVY)
17 July 1968

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRAVELING-WAVE

TYPE 8362

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Forward-wave amplifier, frequency range 5.4 to 5.9 GHz, 50 watts pulse power output, periodic-permanent-magnet focus, convection cooled, coaxial r-f connectors.

PHYSICAL CHARACTERISTICS:

Dimensions:	See Figure 1
Mounting position:	Any
Connectors, r-f:	See Figure 1
Connectors, power:	See Figure 1
Focusing circuit:	PPM integral
Weight:	7 pounds maximum

ABSOLUTE RATINGS: See notes 1 and 2

Parameter:	Ef	If	Ec	egy	Ew	ib	ic
Unit:	V	A, surge	V	v	V	ma	ma
Maximum:	6.6	0.75	-75	475	5750	225	20
Minimum:	6.0	---	-450	---	---	---	---
TEST CONDITIONS:							
(1):	6.3	---	-250	Note 11	Note 11	---	---
(2):	0	---	0	0	0	---	---

ABSOLUTE RATINGS:

Parameter:	Pi(rf)	Du	tp	tk	TA	TE
Unit:	dBm	---	usec Note 3	sec	°C	°C Note 4
Maximum:	27	0.012	50	---	75	150
Minimum:	---	---	---	60	-55	-55
TEST CONDITIONS:						
(1):	Note 25	0.006	26	---	20-30	---
(2):	0	---	---	---	---	---

GENERAL:

Qualification-Required

Marking (see note 5)

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Method	Requirement or Test	Conditions	Symbol	Limits		Unit
				Min	Max.	
	Qualification Inspection Note 28					
---	Temperature	Test condition (1): F=5.9 GHz Note 6	---	---	---	---
---	Low-temperature turn-on	Test condition (1): F=5.9 GHz Note 7	---	---	---	---
---	Humidity	Test condition (1): F=5.9 GHz Note 8	---	---	---	---
---	Humidity turn-on	Test condition (1) Note 9	---	---	---	---
---	Initial Life Test	Test condition (1): F=5.9 GHz, Notes 17, 24, 29	---	---	---	---
	Quality Conformance Inspection, Part 1 Note 28					
1301	Heater current	Ef=6.3 V	If	0.35	0.50	A
1256	Cathode current	Test condition (1): Pi(rf)=0	ib	110	190	mA
---	Grid cut-off	Test condition (1): Pi(rf)=0; egy=0	ib	---	100	uA
4250	Large-signal power output	Test condition (1): F=5.4 to 5.9 GHz Notes 10, 13, 21, 26	pso	17.6	20.0	dBw
			Pi(rf)	7.0	20.0	dBm
			po	17.0	20.0	dBw
			Δpo	---	2.0	dB
			EW	5000	5500	V
			egy	330	430	V
4261	Spurious power output	Test condition (1): Pi(rf)=0; F=4.9 to 6.4GHz Note 14	spo	---	0	dBm
	Quality Conformance Inspection, Part 2 Notes 16, 19					
---	Isolation	Test condition (1): egy=0; F=5.4, 5.9 GHz	Li	70	---	dB
4266	Direct interelectrode capacitance	Test condition (2) Note 15	Cc-all	---	55	pF

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Method	Requirement or Test	Conditions	Symbol	Limits		Unit
				Min.	Max.	
	Quality Conformance Inspection, Part 2 Notes 16, 19 (Cont'd)					
4250	Power output variation	Test condition (1): F=5.4, 5.9 GHz				
		Ew=±2% Note 12	Δpo	---	3.0	dB
		egy=±2% Note 22	Δpo	---	1.25	dB
		Ef=±5% Note 23	Δpo	---	1.5	dB
		Note 13				
---	Vibration	Test condition (1); F=5.65 GHz Note 17	---	---	---	---
1042	Shock	Test condition (2) Note 18				
	Quality Conformance Inspection, Part 3 Notes 19, 20					
---	Life test	Test condition (1), Group S F=5.9 GHz Notes 17, 24, 29	t	5000	---	hrs
---	Life test end point	Test condition (1): F=5.9 GHz; Pi(rf)=20 dBm, maximum	po	16.0	---	dBw

NOTES:

1. Typical operating conditions for this device are described by Test condition (1). Because of the number of interrelated conditions possible with traveling-wave tubes, Absolute ratings shall refer to each individual rating with the tube otherwise operating in the above test condition. Note that ratings are not of themselves test conditions or operating conditions.

2. Cathode and one leg of heater shall be connected internally. Shell, helix and collector shall be at ground potential. All voltages are referred to cathode.

3. The grid pulse, tp, shall comply with the requirements of MIL-STD 1311, method 4304. Pulse shape, with tp=26 usec ± 0.75 usec, and with 0.3 usec and 1.5 usec maximum rise and fall times, respectively.

4. TF is the temperature measured at point "A" on Figure 1.

5. Each tube shall be marked with the following information:
   (a) Serial number
   (b) Cathode voltage for large-signal operation
   (c) Grid pulse voltage for large-signal operation
   (d) Input and output r-f connector identification

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6. The temperature test shall be conducted in accordance with the requirements of MIL-E-16400(NAVY), for class 4 equipment. The tube shall operate within the limits specified under Quality conformance inspection, Part 1, Large-signal power output.

7. With the tube in the environment chamber, reduce the chamber temperature to 0±2 degrees Centigrade, and maintain the chamber temperature at this value for at least 24 hours with the tube in the non-operating condition. At the end of the 24-hour period, turn on the tube. The tube shall operate within the limits specified under Quality conformance inspection, Part 1. Large-signal power output, within 15 minutes from turn-on.

8. The humidity test shall be conducted in accordance with the requirements of MIL-E-16400(NAVY), for class 4 equipment. Reference measurements and all subsequent measurements shall consist of the test for Large-signal power output under Quality conformance inspection, part 1.

9. The humidity turn-on test shall be conducted upon completion of the humidity test. With no voltages applied to the tube, the temperature and humidity shall be maintained at 40±2 degrees Centigrade and 90 to 95 percent, respectively, for not less than 96 hours. The tube then shall be turned on and tested as specified under Quality conformance inspection, part 1, Large-signal power output. There shall be no evidence of sparking and the tube shall operate within specified limits within 15 minutes of turn-on.

10. At each frequency there shall be a range of Pi(rf) of at least 8 dB wherein po is within the limits specified.

11. Ew and egy are defined as the voltages which produce specified large-signal power output within the specified limits, and they shall be identical with the values specified in (b) and (c) of note 5.

12. At each of the specified frequencies adjust Pi(rf) to obtain saturated power output, pso. Reduce Pi(rf) by 4 dB. Vary Ew from Ew-2% to Ew+2% and note power output, po, at each value. Increase Pi(rf) to the saturation value plus 4 dB and repeat the power measurements. The change in power, Δpo, at each value of Pi(rf) shall be within the limits specified.

13. A filter shall be inserted between the tube and the power-measuring device to attenuate harmonic-frequency components by at least 30 dB relative to the power at the fundamental frequency.

14. The input termination shall have a nominal value of 50 ohms, and the output shall be terminated in a variable-phase short circuit. A preselector having a pass band of not more than 30 MHz and a tuning range of 4.9 to 6.4 GHz shall be used between the tube and the power-measuring device. The spurious power output, spo, shall be not greater than the specified value in any portion of the above frequency range at all phases of the short circuit.

15. Cc-all is the capacitance between the grid and all other elements, including the shell. The grid lead shall be at maximum separation from the shell and from all other leads.

16. The Quality conformance inspection, Part 2, shall be performed in accordance with MIL-STD-105, Inspection level 2, with an AQL of 4%. In the event of lot rejection, tightened inspection shall be invoked. Normal inspection shall be resumed when two consecutive lots have conformed with the tests of Quality conformance inspection, Part 2. For inspection purposes the lot size shall be one month's production.

17. The tube shall be rigidly mounted to the vibration table and shall be vibrated in accordance with MIL-STD 167(SHIPS), Type 1, Environmental vibration. The detected output of the tube shall be displayed on an oscilloscope. Prior to vibration, the oscilloscope shall be calibrated in the d-c mode, using slow sweep, by means of a precision attenuator. The face of the display tube shall be marked with lines corresponding to the saturated power output level, and to levels of ±0.5 dB therefrom. The attenuator then shall be returned to the saturated power output level. The displayed variations attributable to vibration shall be within the calibration lines.

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18. The shock test shall be conducted in accordance with MIL-STD 202, method 213, Shock (Specified pulse), using the sawtooth shock pulse, Figure 213-2, with peak acceleration of 40 g, and duration of 10 msec. The tube thereafter shall be tested for Quality conformance inspection, Part 1, Large-signal power output, and shall operate within the specified limits.

19. Tubes for these tests shall be selected at random from the production lot, and shall previously have passed the tests of Quality conformance inspection, Part 1.

20. Tubes selected for this test shall previously have passed the tests for shock and vibration.

21. Adjust Pi(rf) to obtain saturated power output. Using this value of Pi(rf), measure po continuously across the frequency range 5.4 to 5.9 GHz. The total variation in power output, Δpo, across the band shall be not greater than the limit specified herein.

22. At each of the specified frequencies adjust Pi(rf) to obtain saturation. Reduce Pi(rf) by 4 db. Vary egy from egy-2% to egy+2% and note the power output at each value. Increase Pi(rf) to the saturation value plus 4 dB and repeat the power measurements. The change in power, Δpo, at each value of Pi(rf) shall be within the specified limits.

23. At each of the specified frequencies adjust Pi(rf) to obtain saturation. Reduce the heater voltage to the lower of the specified values and read the power output, po. Increase the heater voltage to the higher of the specified values and read the power output, po. Allow at least one minute for stabilization at each value of Ef. The total change in power shall be within the specified limit.

24. The tube shall be operated intermittently according to the following schedule for the duration of the life test:

Heater voltage, Ef. only	4 hours
Test condition (1)	8 hours

25. Pi(rf) is the incident power to the input r-f terminal of the tube. Under test condition 1, unless otherwise specified, Pi(rf) shall be that rf input power required to obtain saturation power output, Pso, at the specified frequency.

26. The tube shall meet the power-output conditions at all frequencies within the band when operating into a load of 1.5:1 VSWR, minimum.

27. Three tubes are required for these tests. Quality conformance inspection, part 1, shall be performed prior to and upon completion of Qualification inspection.

28. All tubes shall be tested in accordance with Quality conformance inspection, part 1.

29. Initial Groups life test shall be conducted in the following sequence:

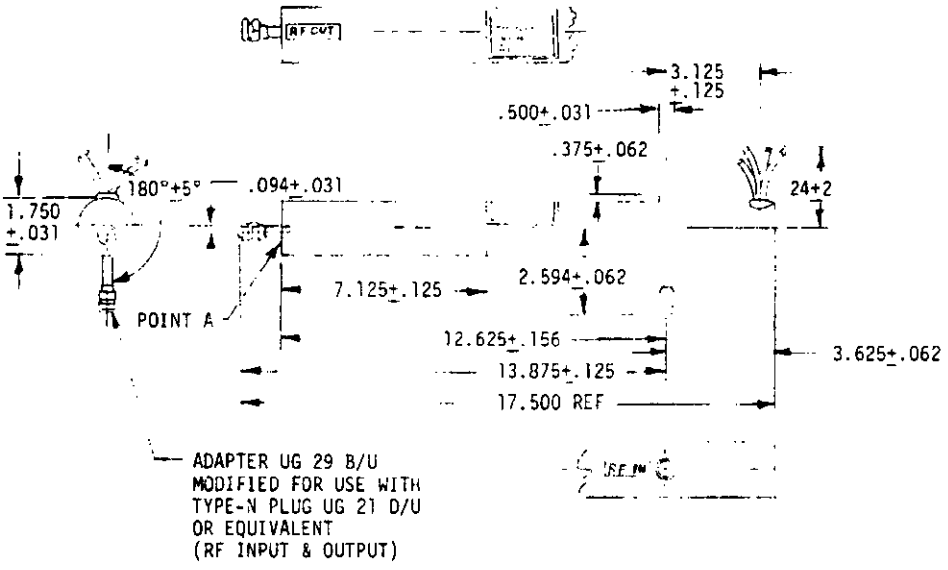
    (a) Operate in accordance with QCI, part 3 for 336±36 hours.
    (b) Test in accordance with QCI, part 1 (less Dimensions), and QCI, part 2, Power output variation.
    (c) Operate in accordance with QCI, part 3 for 72 hours at F=5.9GHz and simultaneously operate in accordance with QCI, part 2. Paragraph 3.1.4.3.3 of MIL-STD 167(SHIPS), "Endurance test" shall apply. The vibration frequency shall be the resonant value determined from QCI, part 2, Vibration test, or 33Hz, with amplitude as shown in MIL-STD 167, Table 1.

User activities:
Navy-AS, OS, MC, CC, SH

Preparing activity:
Navy - EC

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LEAD COLOR CODE

Yellow --- Heater-Cathode
Brown --- Heater
Green --- Control Grid
Black --- Shell-Ground

NOTE

Point - A = Location where envelope temperature is measured

FIGURE 1. Outline drawing of electron tube type 8362.

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